KisanNiti: Cooperative Post-Harvest Selling for Farmer Producer Organizations with Mandi Price Forecasting, Storage Sensing and Fair Allocation

Harkiran Kaur, Arijit Singh, Tushar Garg, Hitesh Mehandiratta, and Namya Gautam

Thapar Institute of Engineering and Technology, Patiala, Punjab, India
{asingh28_be23,tgarg3_be23,hhitesh_be23,ngautam_23}@thapar.edu

Abstract. A Farmer Producer Organization (FPO) must decide where, when and how much of its members' produce is sold, and existing tools treat forecasting, storage and logistics separately. KisanNiti connects them in five stages that a manager follows: collecting and cleaning prices and farmer records, proposing a plan, reviewing it, telling each farmer in their language, and recording what was sold. A regime test separates price-bound from market crops: for Paddy (Common) in Punjab, at least 96.1% of reported prices in every season from 2020 to 2025 equal the minimum support price, and a rule predicting it had the lowest error (Rs 1.26 per quintal). For basmati, eight forecasting methods were compared under a protocol frozen before any test result existed; on the held-out season none of 42 comparisons with the naive forecast at 1, 7 and 14 days found a method better after correcting for multiple comparisons. An exact fair-allocation model raised the smallest share a farmer sells from 0.0% under the nearest-mandi rule to 68.6% in 60 synthetic plans at real prices, at Rs 5.70 per quintal. The system was built and tested end to end (1,372 automated tests, a 14-step run on fictional data); no real farmer data or user study is included.

Keywords: Farmer Producer Organizations · Mandi price forecasting · Minimum support price · Fair allocation · Integer programming

1 Introduction

A Farmer Producer Organization (FPO) is a registered body of farmers that can pool, market and sell its members' produce [23]. For each member lot the FPO must choose a mandi, a day and a quantity, and the choices depend on each other: the best-paying mandi may be the farthest, a later date keeps grain in storage longer, and when many members deliver in one week a mandi can receive more than it can handle in a day. Post-harvest losses in India are large, about Rs 1.53 trillion of food a year by a 2024 ICRIER estimate that names storage as one cause [12].

Two public sources give an FPO most of the price information it needs. AGMARKNET publishes daily arrivals and the minimum, maximum and modal price reported by each mandi [9], and the government announces a minimum support price (MSP) for paddy and wheat before each season [8,7]. The two cases differ sharply. For Paddy (Common) in Punjab, at least 96.1% of the reported prices in every season from 2020 to 2025 were exactly the Grade-A MSP in the 20 mandis studied. Basmati paddy has no MSP and is

sold at market prices that differ by variety, so for basmati the mandi and the day can change what a farmer earns.

KisanNiti is a decision-support system for FPO staff in the Patiala area. It reads AGMARKNET prices, the FPO's lot records and a grain-storage sensor, proposes a selling plan, lets a manager review it, and tells each farmer in their language (Fig. 1). The paper makes five contributions. A regime test, the share of reports equal to the MSP, separates price-bound from market crops (Sect. 4.1). For basmati, a comparison of eight forecasting methods with prediction ranges was fixed in a pre-registered protocol and evaluated once on a held-out season (Sect. 4.2). An exact allocation model shares mandi limits fairly among members and is compared with simpler rules (Sect. 4.3). A storage-sensing rule is tested on simulated scenarios (Sect. 4.5; details in the supplement). The five stages from data to recorded sales are built and tested as one system (Sect. 4.6). Claims are limited to what each experiment supports; Sect. 5 lists what is not shown.

2 Related Work

Price forecasting. Most Indian studies take daily or monthly AGMARKNET prices and compare one model family with ARIMA. The largest comparison covers 23 commodities in 165 markets from 2010 to 2024, where LSTM and GRU models had the lowest errors [18]; adding rainfall and temperature helped deep models on tomato, onion and potato [20]. Closer to the crops here, Bhoi et al. fitted six time-series models to monthly basmati prices from 74 Punjab markets and reported forecasts as probability bands [5], and Balantrapu et al. asked what it means for a series to be anchored to the MSP: for wheat, a naive forecast was almost as good as their best model [4]. Graph models forecast many related series at once [22,19]; T-GCN and A3T-GCN were built for traffic [25,3].

Storage and sensing. Measured farm storage losses in North India are small for the first months: 0.39 to 0.50% for wheat in Madhya Pradesh and zero for paddy in Punjab, where farmers sell soon after harvest, with losses expected to rise after about three months [11]. Low-cost monitors log temperature and humidity and raise alarms [15], but none of these systems passes its readings to a selling decision.

Allocation and fairness. Mixed-integer models already plan agricultural logistics for small growers, cooperatives and food-grain freight [24,1]. They minimise cost or maximise total profit and do not say how gains are shared. Leximin (lexicographic max-min) methods exist for that [21,6] and have been used to allocate school classrooms with real data [16]; no agricultural allocation paper read for this study used a fairness criterion.

Advice to farmers. Phone advice has a mixed record: SMS price and weather messages to farmers in 100 Maharashtra villages produced no significant change in prices received [10].

Table 1 lists only what each paper reports. In the 38 studies reviewed, price forecasting, storage monitoring and allocation were studied separately. No study fed a forecast into a plan that shares mandi capacity among the members of an FPO, and none combined such a plan with a fairness rule or a manager's approval before farmers are told.

Table 1. Features reported by representative studies and built in KisanNiti (Y: reported; –: not reported in what was read). D: Indian mandi data; M: MSP check; R: forecast ranges; H: hold-or-sell with storage cost; S: storage sensing in the decision; C: shared mandi capacity; F: fairness criterion; A: manager approval; W: farmer messages. KisanNiti ranges come from baseline forecasts; hold-or-sell is not built.

Study	D	M	R	H	S	C	F	A	W
Manogna et al. [18]	Y	–	–	–	–	–	–	–	–
Bhoi et al. [5]	Y	–	Y	–	–	–	–	–	–
Balantrapu et al. [4]	Y	Y	–	–	–	–	–	–	–
Anoop et al. [1]	–	–	–	–	–	Y	–	–	–
Kurokawa et al. [16]	–	–	–	–	–	–	Y	–	–
Fafchamps and Minten [10]	–	–	–	–	–	–	–	–	Y
KisanNiti v2 (tested)	Y	Y	–	–	–	Y	–	–	–
KisanNiti (this paper)	Y	Y	Y	–	Y	Y	Y	Y	Y

KisanNiti addresses this combination for Punjab, where basmati is the main crop sold at market prices and nearly all reported prices of common paddy equal the MSP.

3 System Design and Methods

KisanNiti follows the order in which an FPO manager works (Fig. 1): (1) farmer records, mandi prices and storage readings are collected and cleaned; (2) a forecast, a storage assessment and a fair allocation produce a proposed plan; (3) the manager reviews, edits and approves or rejects it; (4) each farmer is told in their own language; and (5) replies and actual sales are recorded and compared with the plan. Five rules hold throughout: raw facts, derived values, proposals, human decisions and what actually happened are stored separately and never overwrite each other; quantities and money are exact (grams and paise, never floating point); each FPO sees only its own data, enforced in the query and again in the database; every change is an append-only record with its author; and every failure is explicit, so a missing contact, consent or language blocks a message with a stated reason instead of being guessed.

3.1 Stage 1: Data and Cleaning

Prices come from the AGMARKNET report *Date Wise Prices for Specified Commodity*, downloaded by hand each month because the portal asks for a CAPTCHA. 29 months of Paddy (Basmati) for Punjab are filed with checksums. A parser keeps every line with its source file and position, so each cleaned row traces to its source; the files hold 6,678 rows (6,098 dated, 580 without a date) and every row is accounted for. Mechanical checks flag suspicious rows but remove none. Where the portal shows a variety under the date of the row above, the date is inferred and flagged. Prices of one variety become a daily index, the median modal price over the mandis that reported the variety that day (at least three, no filling); it is a market summary, not the price a farmer received. Farmer records (members, lots with exact quantity, an append-only stock ledger) are entered by staff, and sensor readings are assessed hourly (Sect. 3.3). The cleaned release is imported with an

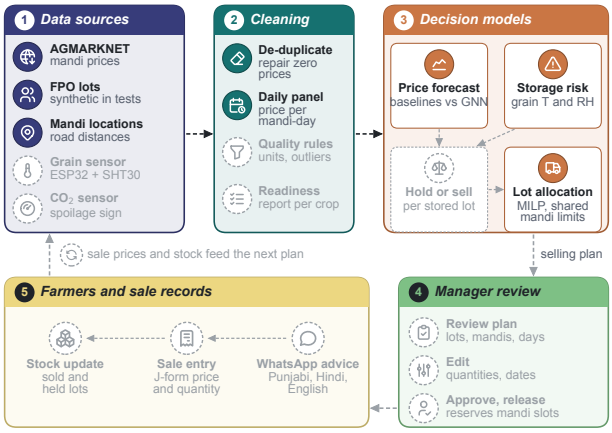


Fig. 1. KisanNiti architecture. Data sources (1) are cleaned into a daily price panel (2). Decision models (3) turn prices, storage readings and lot records into a plan that the FPO manager reviews (4). Farmers receive the approved plan on WhatsApp, and recorded sales update stock for the next plan (5). MILP: mixed-integer linear program; GNN: graph neural network; RH: relative humidity.

exact row reconciliation and the status `PENDING_HAND_CHECK`; it becomes `VALIDATED` only after a hand check of 56 stratified rows against the portal, which has not been done, so this paper reports the data as unchecked.

3.2 Stage 2: Forecasting, Storage Risk and Fair Allocation

Regime test. For each crop and season, the share of reported prices equal to the announced MSP is computed. A high share marks a price-bound crop, for which the MSP is used as the price and no forecaster is trained.

Pre-registered forecast comparison. For basmati (1121 and 1509) the index is forecast at 1, 7 and 14 calendar days (28 secondary, 60 descriptive). Seasons 2021–22 to 2023–24 give the initial training, 2024–25 is used to tune and calibrate, and 2025–26 is held out and evaluated once; September and October 2026 are a separate partial check. Each forecast uses only data up to its origin day, and an origin needs ten index days of its season. Eight methods were fixed beforehand: the last value, a seasonal naive value, a five-day average, ETS, ARIMA, LightGBM, a small GRU and the pretrained Chronos-2 model [2] used zero-shot, with 80% and 95% ranges from analytic intervals, quantile objectives, split-conformal residuals or the model's quantiles. Accuracy is the MAE in rupees per quintal and MASE. Each model is compared with the naive value by a Diebold–Mariano test with the Harvey–Leybourne–Newbold correction, Holm-corrected over all 42 comparisons, with a block-bootstrap interval of the relative change in MAE. A model counts as better only if the adjusted $p < 0.05$, the interval excludes zero and the gain exceeds the smallest the sample can detect (13.3% for 1121, 17.2% for 1509); otherwise

the product keeps the naive value. The protocol and its choices were frozen before the run, the code is tested on synthetic series, and a gate in the runner refuses any run before the freeze, on changed code or data, or a second test run.

Fair allocation. A plan gives each lot a mandi, a day and a quantity, and each mandi has a daily limit shared by all members. An exact mixed-integer program with a fixed hierarchy allocates as much as possible, then maximises the smallest share any farmer receives (and so on, lexicographically), then gives lots at risk of spoilage the earliest dates, then maximises net return, then breaks ties deterministically. An independent checker re-validates every plan, and a brute-force reference confirms the solver on small cases.

3.3 Storage Sensing and Risk Rule

One protected SHT30 temperature and humidity probe (SHT30) is placed in a bag of grain and read every five minutes by an ESP32 board; readings are sent by MQTT with TLS, kept in a 24-hour buffer until the server confirms it has stored each one once, and assessed hourly. Grain balances its moisture with the air between the kernels, and that air's relative humidity divided by 100 is its water activity a_w. Storage fungi need at least 65 % relative humidity [13], and grain below 65 to 70 % degrades slowly [17]. The rule takes the median humidity and temperature of the last 24 h and returns LOW for $a_w < 0.65$, MEDIUM for $0.65 \leq a_w < 0.70$ and HIGH from 0.70; the level rises at once but falls only after 48 h at least 2 points below the limit; it returns UNKNOWN, with reasons, when the probe hangs in shed air, in the first 6 h, or when readings are old or missing. It says whether the air in a bag favours mould and insects; it does not measure moisture content or detect spoilage.

3.4 Stages 3 to 5: Review, Messages, Sales

A proposed plan is stored unchanged. A manager's edit becomes a new revision, accepted only if the independent validator finds no violation. Approval needs the approve permission and an acknowledgement of every warning, and in one database transaction records the decision and reserves exact quantities in the stock ledger; a reservation cannot exceed free stock, approval cannot be given twice, and two plans competing for one lot cannot both succeed (tested with parallel requests on PostgreSQL). After approval, one message per farmer is built from a versioned template in the farmer's language (Punjabi, Hindi or English) and the approved facts only, with no price and no promise. A message needs an active farmer with a phone number and consent to WhatsApp messages, rechecked at sending; a preview channel shows the exact text and sends nothing. Without a messaging-platform account, the manager presses "Send to farmers", confirms with yes or no, and receives one link per farmer that opens his own WhatsApp with the text written; he sends it there and reports "sent" or "not sent". The system cannot see delivery, so that report is stored as his statement, never as a delivery. Replies are entered by staff and never read as acceptance. Sales are entered from the paper J-form with exact amounts (ties to even), duplicate form numbers refused and mistakes voided by compensating records; a sale on plan consumes its reservation, and an off-plan sale needs the person to choose which reservation to give up (a provisional rule). The system reports planned against actual quantity, price and share per farmer.

Table 2. Paddy (Common) prices in the 20 selected Punjab mandis, September to December of each season. SD: standard deviation.

Season	MSP Grade A (Rs/q)	Trading dates	Prices equal to MSP (%)	SD of prices (Rs/q)
2020	1,888	70	98.3	44.3
2021	1,960	55	99.1	80.5
2022	2,060	60	98.5	16.5
2023	2,203	70	96.1	45.5
2024	2,320	66	98.8	29.5
2025	2,389	61	99.0	21.6

4 Results and Discussion

4.1 Paddy (Common): Prices Stay at the MSP

Table 2 describes the 20 Punjab mandis that reported in all six seasons: 4,626 mandi-days over 382 trading dates (60.5% of those possible). In 2024, 98.8% of the reported prices were exactly the Grade-A MSP of Rs 2,320 per quintal; the lowest share in any season was 96.1% in 2023, and in 2024 the mean daily standard deviation across mandis was only Rs 0.4.

Seven forecasters were compared on the three latest seasons at horizons of 1 to 7 trading dates: the MSP rule (the announced Grade-A MSP), the last reported price, a seven-date average, ridge regression, a GRU, T-GCN and A3T-GCN. The MSP rule had the lowest mean absolute error, Rs 1.26 per quintal, ahead of the last price (Rs 1.77); the best learned model, A3T-GCN, reached Rs 2.36, ahead of the GRU (Rs 2.51) and T-GCN (Rs 2.54). At one trading date A3T-GCN was worse than the last price on 158 of 158 dates: a mandi that stays at the MSP is predicted exactly by the last price, and a network almost never predicts a change of exactly zero. The MSP rule and the last price are close (better on 13 dates, worse on 19, equal on 126), and no significance test was run on this comparison. For Paddy (Common) the announced MSP is therefore used as the price and no forecaster is trained.

4.2 Basmati Paddy: the Pre-Registered Forecast Comparison

The comparison was run once on the held-out season after the protocol and the configuration were frozen (commits `C.0 freeze` and `C.0 configuration freeze` in the repository), on the dated-only index; the hand check of the blank-date rule is still open and cannot change the primary result. The index has 433 days for 1121 and 369 for Basmati 1509, because index days exist only from September to January. After the rule that an origin needs ten index days of its season, the held-out season offers 68, 64 and 60 pairs at 1, 7 and 14 days for 1121 and 62, 57 and 51 for 1509, fewer than the 60 first planned for 1509, and 38 at 28 days, below the 40 pairs needed for a coverage claim. An exploratory power check on earlier seasons puts the smallest detectable improvement over the naive forecast at 11 to 17% of its error, so “the naive forecast is hard to beat” is a valid outcome.

Table 3. Held-out season 2025–26: MAE (Rs per quintal) at 1, 7 and 14 days for 1121 and Basmati 1509, and change in MAE against the naive forecast (%). No change was judged better by the pre-registered rule.

MAE	1121			Basmati 1509		
Method	1 d	7 d	14 d	1 d	7 d	14 d
Naive	107	107	144	62	112	148
Seasonal naive	225	209	194	247	240	223
Five-day average	87	111	160	81	98	160
ETS	95	108	159	62	111	148
ARIMA	100	122	178	65	115	162
LightGBM	100	141	264	71	123	147
GRU	80	106	168	76	104	167
Chronos-2	89	123	186	65	125	170
Change vs naive (%)						
Seasonal naive	+110.7	+100.4	+34.8	+296.7	+113.5	+50.6
Five-day average	-18.7	+3.3	+11.2	+30.5	-12.7	+7.6
ETS	-10.7	+1.1	+10.3	-1.0	-1.7	-0.2
ARIMA	-6.2	+13.7	+23.6	+3.9	+2.3	+9.1
LightGBM	-5.9	+31.5	+83.7	+14.3	+9.4	-0.8
GRU	-25.3	-1.0	+17.0	+22.7	-7.3	+12.4
Chronos-2	-16.7	+14.6	+29.2	+4.6	+11.4	+14.8

Cleaning choices were measured before any forecast was run. Four outlier screens (past-only median absolute deviation with $k = 3$ and $k = 5$, a Hampel filter, an interquartile rule) flag 2.2% to 8.8% of the dated rows, and excluding them would change 27 to 110 of the 433 index days of 1121, by up to Rs 770 per quintal; the hand check must decide which screen is right. As a plausibility check of the inferred dates (weaker than the hand check), the 486 same-day pairs of 1121 and 1509 in one market, whose second row always has an inferred date, have a median price ratio of 1.18 against 1.17 for 211 own-dated pairs one day apart, with the smaller spread expected of same-day pairs (interquartile range 0.14 against 0.17); only 2 of 519 inferred rows fall on a key that already has another row.

Under the pre-registered rule, none of the 42 comparisons found a method better than the naive value (0 better). The largest raw improvement was the GRU at one day for 1121, a change in MAE of -25.3% (Rs 80 against Rs 107); its bootstrap interval excluded zero, but the Holm-adjusted p was 0.252. The only methods that differed significantly from the naive value were worse: the seasonal naive forecast was significantly worse in 3 comparisons (Basmati 1509 at one day: change +296.7%, adjusted p 0.000), because one season's price level says little about the next. The naive value is therefore a strong forecast for prices that move this little from day to day, and the promotion rule keeps it: the product uses the naive value with ranges and states that the baseline is enough.

Secondary and descriptive horizons are reported without a test. At 28 days the GRU and LightGBM had a lower MAE than the naive value for 1121 (Rs 173 for the GRU against Rs 278), as in validation; both use the day of the season as a feature, so this is a hypothesis for the next season, not a finding. The partly observed season 2026–27 was run separately and is labelled partial: 6 pairs at one day for 1121 and 39 for 1509, falling to 26 at 14 days, so no test was run. For 1509 the naive value had the lowest MAE

Table 4. Selling policies on 60 synthetic plans at real 2025 prices. Valid: plans accepted by the independent validator (a plan is invalid if it exceeds a mandi's daily limit). Net: mean net return (Rs per quintal). Min. share: smallest share of a lot sold, averaged over plans. Fairness price: net return given up against the utilitarian plan.

Policy	Valid (%)	Net	Min. share (%)	Gini	Fairness price	Solve (s)
Nearest mandi	100	2,329.69	0.0	0.3125	14.99	0.00
Independent best	0	2,360.49	100.0	0.0000	–	0.00
Utilitarian	100	2,344.68	0.0	0.3087	0.00	0.07
Max-min only	100	2,339.87	68.6	0.0019	4.81	0.57
Greedy	100	2,330.60	68.6	0.0019	14.08	0.07
KisanNiti	100	2,338.98	68.6	0.0019	5.70	4.94
Actual-price bound	100	2,338.98	68.6	0.0019	5.70	5.07

at 7 and 14 days (Rs 90 and Rs 137), while the GRU and LightGBM were clearly less accurate (Rs 210 and Rs 203 at 7 days).

Ranges at 80% and 95% were assessed at the three primary horizons. The models' own ranges (ETS, ARIMA, LightGBM quantiles, Chronos-2) were calibrated in 37 of 48 cases, too wide in 7 and too narrow in 4. The split-conformal ranges, calibrated on the residuals of the single, more volatile validation season, were calibrated in only 13 of 60 and too wide in 47: one season is too little to calibrate a range.

4.3 Allocation: Fair Plans Against Simple Rules

Before any comparison, the solver was checked against a brute-force enumeration that shares only the input types and the exact objective with it. On 324 random small instances, three back ends (HiGHS, SCIP, CP-SAT) and two leximin methods, 1,728 of 1,728 runs reached exactly the enumeration's complete objective. The method was then compared with simple rules on 60 weekly plans: ten draws of a synthetic 300-member FPO whose lots become ready over six weeks, scored at real 2025 AGMARKNET prices of Paddy (Common) and real mandi coordinates, every plan re-checked by the independent validator. The rules are the nearest mandi, each farmer's independent best choice, a plan that maximises only total net return (utilitarian) or only the smallest share, a greedy water-filling heuristic, the KisanNiti hierarchy, and the same hierarchy with actual prices (a bound).

Independent choices broke a mandi's daily limit in 100% of the plans, so none was a valid plan. The nearest-mandi rule respects the limits but left the least-served farmer with 0.0% of their lot sold, and the utilitarian plan did the same (0.0%), because maximising the total ignores who is left out. The KisanNiti hierarchy raised the smallest share to 68.6%, the most the mandi limits allow, with a Gini coefficient of 0.0019, at Rs 5.70 per quintal against the utilitarian plan. The greedy rule reaches the same smallest share but gives up Rs 14.08 per quintal. Actual instead of forecast prices changed nothing, as expected: prices of Paddy (Common) hardly differ between mandis (Sect. 4.1), so the plan is decided by distance, limits and fairness. Of the 240 exact solves, 240 were proven optimal. Three weighted-sum alternatives were tried in a six-plan pilot; all reached the 300-second limit without a proof and are left out of Table 4. The experiment uses a synthetic FPO and shows properties of the method, not a gain for any real farmer.

The transport rate and the daily limit are invented inputs, so the comparison was repeated for 9 combinations of three rates (Rs 1, 2 and 4 per quintal-km) and three limits (200, 300 and 500 quintals per mandi-day). The KisanNiti plan was valid in every setting. The smallest share it secures rises with the limit, from 45.67% to 98.17%, because no plan can sell more than the mandis can take; its gain over the nearest-mandi rule is Rs 3.81 to Rs 19.19 per quintal and its price of fairness Rs 0.74 to Rs 17.15. The greedy heuristic gave up more than the exact plan in 9 of 9 settings. The nearest-mandi rule left some farmer without any sale whenever the limit was 300 quintals or less. Solve times were measured on random instances of 10 to 1,000 farmers (5 to 25 mandis, 3 to 14 days, a 60-second limit, three seeds each). HiGHS and SCIP proved optimality on every instance up to 100 farmers (median 22.14 and 36.85 seconds at 100) and reached the limit with a feasible plan at 300 and 1,000. CP-SAT proved optimality only at 10 farmers, because its integer scaling becomes too large beyond that, and the greedy heuristic needed under 0.23 seconds even at 1,000. Every plan a solver returned was valid. The 300-member persona above was solved exactly in a mean of 4.94 seconds (maximum 37.6); instance structure matters, and larger or less regular FPOs need a time limit and the greedy plan as a fallback.

4.4 Checking the Manager's Edits

A manager's edit must be accepted exactly when it keeps the plan valid. An independent validator re-checks every revision, and a second checker written directly from the rules (admissible choice, no duplicate line, whole increments, no lot over-allocated or, when the whole lot must be sold, left over, no mandi over its strict limit) was compared with it on 4,431 random edits of 500 solved instances, from harmless changes to a single broken rule (1,052 valid, 3,379 invalid). The two gave the same verdict on 4,431 of them.

4.5 Storage Rule on Simulated Scenarios

The node has not run in real grain, so the rule was tested on 6 simulated scenarios built from the monthly climate normals of the IMD station at Patiala [14], each with 20 noise seeds and sensor biases of ±2 %RH. Without bias it agreed with the intended category in 88–100 % of known hours in the four in-grain scenarios and changed category at most 0.7 times a week; the latest-reading rule changed 1.4–19.1 times, and shed-air humidity was a poor stand-in for the grain. The costs are delay (the 24-hour median reported HIGH about 12 h late) and unknowns. A ±2 %RH sensor error mattered as much as the choice of rule (up to 29 % false HIGH hours), so a salt calibration is required before real readings are interpreted. This measures how far noise and shortcuts move a rule from its definition, not whether the definition predicts spoilage.

4.6 The Built System

The backend, shared algorithms, sensor ingestion and interface contracts hold 1,372 automated tests (including parallel approvals on PostgreSQL, isolation between FPOs, append-only records and exact message texts) and 12 versioned contracts with 137 valid

and 223 invalid examples. A single command runs 14 steps on two new fictional FPOs through the interface the website uses (data, stock, a simulated sensor, plan, edit, approval, messages, the manager's hand-over to WhatsApp, sales, release, a ledger audit and isolation), checks each step against an independent recomputation of amounts, rounding, ledger and message texts, and stops at the first deviation; a full run takes about 18 seconds and passed all 14 steps. A corrupted ledger entry and planted errors (rounding down, a wrong weighted price, a message losing its last character) were each detected. All farmers, lots and sales are fictional; no user study was done, no real message was sent, and the Punjabi and Hindi texts have not been read by a native speaker.

5 Conclusion and Future Work

Selling decisions of an FPO depend on whether the crop's price is set by policy or by the market. For Paddy (Common) in Punjab it is set by policy: the announced support price was the most accurate forecast, no learned model beat the last price, and the useful work is to share mandi limits fairly. The exact allocation raised the smallest share a farmer receives from 0.0% to 68.6% in the synthetic experiment and recovered part of what a greedy rule gives up. For basmati the forecast comparison was fixed in advance, and its result is negative: on the held-out season no method of eight beat the naive value at 1, 7 or 14 days after correcting for multiple comparisons (Sect. 4.2), so for the product the baseline is enough. The five stages from data to recorded sales were built as one system and tested end to end, including an approval that reserves stock exactly and a way for the manager to send each farmer the message from his own WhatsApp.

Limits. The index is a market summary, not the price a farmer received, and exists only from September to January over five seasons, so only large improvements can be detected. The cleaned prices have not passed the hand check. The FPO in the allocation experiment is synthetic, so no income gain is claimed. The storage sensor was tested on simulated scenarios only. Held wheat is not forecast because Punjab mandis report too few wheat prices outside procurement. The WhatsApp hand-over cannot see delivery, the Punjabi and Hindi texts have not been read by a native speaker, off-plan sales follow a provisional rule, and no user study has been done.

Future work. A one-week sensor run in grain; a hold-or-sell rule with ranges evaluated per season; the allocation comparison with basmati prices and real transport costs; a deployment with one FPO after ethics approval, native-speaker review of the messages and a legal review of the privacy controls; and the official messaging channel, so that one action reaches every farmer.

Acknowledgments. No funding is declared for this work.

Use of AI tools. Claude (Anthropic), a large language model, was used to draft the text from the result files, to write the analysis, figure and table code, and to run consistency checks. Every number in the text is generated by a script from a results file, and the scripts that re-check the stated numbers and the bibliography are in the repository. The authors are responsible for the content of this paper.

Data and code availability. Code, result files and the pre-registered protocol with its commit history are in the project repositories. AGMARKNET prices are public; all farmers, lots and sales in the system experiments are synthetic.

Disclosure of Interests. No competing interests are declared that are relevant to the content of this article.

References

1. Anoop, K.P., Panicker, V.V., Siby, J., Aryadutt, C.S.: Multi-objective optimization algorithms for freight allocation in a food grain supply chain. Applied Soft Computing **171**, 112729 (2025). https://doi.org/10.1016/j.asoc.2025.112729
2. Ansari, A.F., Stella, L., Turkmen, C., Zhang, X., Mercado, P., Shen, H., Shchur, O., Rangapuram, S.S., Pineda Arango, S., Kapoor, S., Zschiegner, J., Maddix, D.C., Wang, H., Mahoney, M.W., Torkkola, K., Wilson, A.G., Bohlke-Schneider, M., Wang, Y.: Chronos: learning the language of time series. arXiv preprint arXiv:2403.07815 (2024). https://doi.org/10.48550/arXiv.2403.07815
3. Bai, J., Zhu, J., Song, Y., Zhao, L., Hou, Z., Du, R., Li, H.: A3T-GCN: attention temporal graph convolutional network for traffic forecasting. ISPRS International Journal of Geo-Information **10**(7), 485 (2021). https://doi.org/10.3390/ijgi10070485
4. Balantrapu, S.P., Raju, G.R., Mudigonda, S.S., Kalur, L.A.: Forecasting minimum support price-anchored mandi prices for wheat and maize in India: a comparative machine-learning and time-series framework. Agricultural Science Digest (2026). https://doi.org/10.18805/ag.D-6567
5. Bhoi, P.B., Vatta, K., Kumar, S., Arora, K., Adhale, P., Vani, G.K.: Probability analysis for forecasting basmati prices in Punjab: application of advanced forecasting models. Journal of Agricultural Development and Policy **33**(2), 157–164 (2023). https://doi.org/10.63066/23220457.33.2.005
6. Chen, V.X., Hooker, J.N.: A guide to formulating fairness in an optimization model. Annals of Operations Research **326**(1), 581–619 (2023). https://doi.org/10.1007/s10479-023-05264-y
7. Directorate of Economics and Statistics, Ministry of Agriculture and Farmers Welfare: Minimum support price (marketing season-wise), as on 28.05.2025, https://desagri.gov.in/wp-content/uploads/2025/08/MSP-Statement_English.pdf, last accessed 2026/10/22
8. Directorate of Economics and Statistics, Ministry of Agriculture and Farmers Welfare: Minimum support prices (according to crop year), as on 22.01.2025, https://desagri.gov.in/wp-content/uploads/2025/02/MSP_for-website_English-version.pdf, last accessed 2026/10/22
9. Directorate of Marketing and Inspection, Ministry of Agriculture and Farmers Welfare: AGMARKNET – agricultural marketing information network, https://agmarknet.gov.in, last accessed 2026/10/22
10. Fafchamps, M., Minten, B.: Impact of SMS-based agricultural information on Indian farmers. The World Bank Economic Review **26**(3), 383–414 (2012). https://doi.org/10.1093/wber/lhr056
11. Gulati, A., Das, R., Winter-Nelson, A.: Reducing Post-Harvest Losses in India: Farmer-Level Interventions and Grain Management Strategies. Indian Council for Research on International Economic Relations (ICRIER), New Delhi (2024), https://icrier.org/pdf/Reducing_Post-Harvest_Losses_in_India.pdf, last accessed 2026/10/22

12. Gulati, A., Das, R., Winter-Nelson, A.: Reducing post-harvest losses in indian agriculture: A case study of selected crops. Policy Brief 20, Indian Council for Research on International Economic Relations (ICRIER), New Delhi (2024), https://icrier.org/pdf/Policy_Brief_20.pdf, last accessed 2026/10/22
13. Gwinner, J., Harnisch, R., Mück, O.: Manual of the Prevention of Post-Harvest Grain Losses. Deutsche Gesellschaft für Technische Zusammenarbeit (GTZ) GmbH, Post-Harvest Project, Eschborn, 2 edn. (1996), https://www.fao.org/4/x5065e/x5065E00.htm, chapter 6, Fungi and mycotoxins. Last accessed 2026/10/22
14. India Meteorological Department: Climatological tables of observatories in India 1991–2020. Tech. rep., Climate Research and Services, India Meteorological Department, Pune (2022), https://imdpune.gov.in/library/public/Climatological%20Tables%201991-2020.pdf, ninth edition. Station Patiala (42101): PDF pages 635–636; the twelve monthly 17:30 IST humidity means were checked against the table on 2026/10/28. Last accessed 2026/10/28
15. Islam, M.S., Sadmani, S., Ali, M.R., Sayem, N.S., Islam, M.H., Hanif, M.A., Alam, M.M.: Remotely controlled smart monitoring system of hermetic paddy storage to reduce postharvest losses in Bangladesh. Smart Agricultural Technology **8**, 100468 (2024). https://doi.org/10.1016/j.atech.2024.100468
16. Kurokawa, D., Procaccia, A.D., Shah, N.: Leximin allocations in the real world. ACM Transactions on Economics and Computation **6**(3–4), 1–24 (2018). https://doi.org/10.1145/3274641
17. de Lucia, M., Assennato, D.: Agricultural Engineering in Development: Post-Harvest Operations and Management of Foodgrains. No. 93 in FAO Agricultural Services Bulletin, Food and Agriculture Organization of the United Nations, Rome (1994), https://www.fao.org/4/t0522e/T0522E00.htm, last accessed 2026/10/22
18. Manogna, R.L., Dharmaji, V., Sarang, S.: Enhancing agricultural commodity price forecasting with deep learning. Scientific Reports **15**, 20903 (2025). https://doi.org/10.1038/s41598-025-05103-z
19. Min, Y., Kim, Y.R., Hyon, Y., Ha, T., Lee, S., Hyun, J., Lee, M.R.: RNN and GNN based prediction of agricultural prices with multivariate time series and its short-term fluctuations smoothing effect. Scientific Reports **15**, 13681 (2025). https://doi.org/10.1038/s41598-025-97724-7
20. Nayak, G.H.H., Alam, M.W., Singh, K.N., Avinash, G., Kumar, R.R., Ray, M., Deb, C.K.: Exogenous variable driven deep learning models for improved price forecasting of TOP crops in India. Scientific Reports **14**, 17203 (2024). https://doi.org/10.1038/s41598-024-68040-3
21. Ogryczak, W., Luss, H., Pióro, M., Nace, D., Tomaszewski, A.: Fair optimization and networks: a survey. Journal of Applied Mathematics **2014**, 612018 (2014). https://doi.org/10.1155/2014/612018
22. Özden, C., Bulut, M.: Spectral temporal graph neural network for multivariate agricultural price forecasting. Ciência Rural **54**(1), e20220677 (2024). https://doi.org/10.1590/0103-8478cr20220677
23. Small Farmers' Agribusiness Consortium: Frequently asked questions: Farmer producer organisations, https://sfacindia.com/fpofaq.aspx, last accessed 2026/10/22
24. Urra-Calfuñir, B.A., Monardes-Concha, C.A., Miranda-González, P.A.: Agri-food supply chain optimization through a decentralized production process in the olive oil industry. Computers & Industrial Engineering **192**, 110185 (2024). https://doi.org/10.1016/j.cie.2024.110185
25. Zhao, L., Song, Y., Zhang, C., Liu, Y., Wang, P., Lin, T., Deng, M., Li, H.: T-GCN: a temporal graph convolutional network for traffic prediction. IEEE Transactions on Intelligent Transportation Systems **21**(9), 3848–3858 (2020). https://doi.org/10.1109/TITS.2019.2935152